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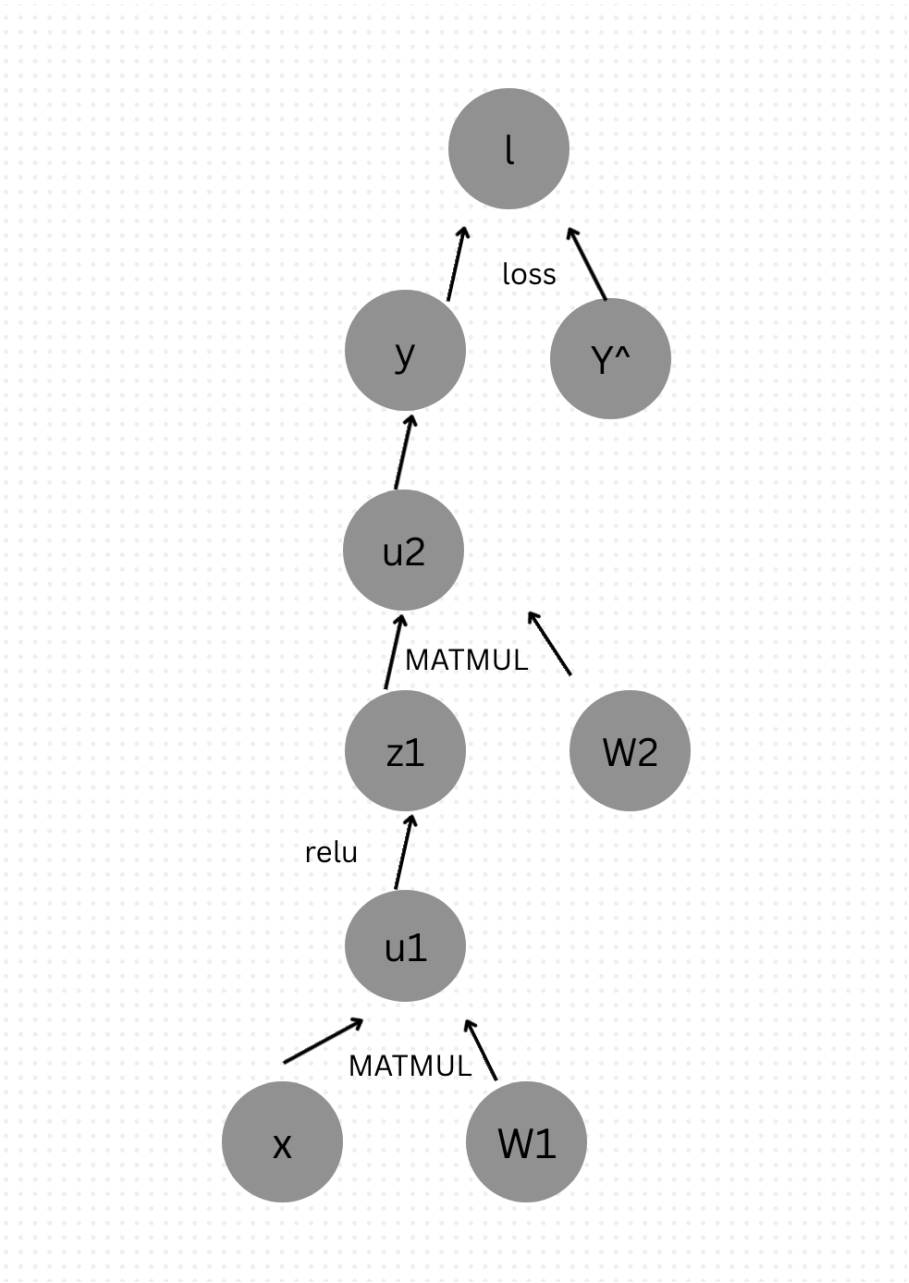
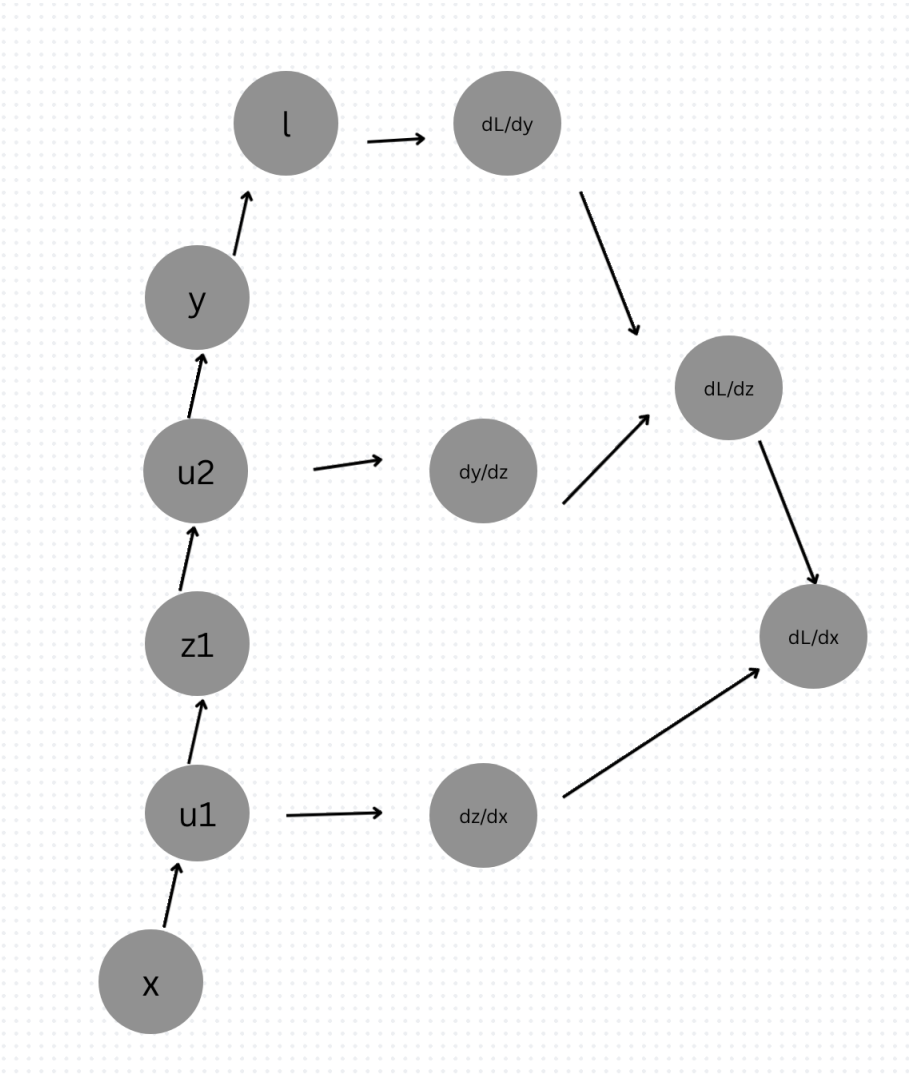
Assignment Sheet Nr. 1

Question 1: Symbol to Symbol derivatives

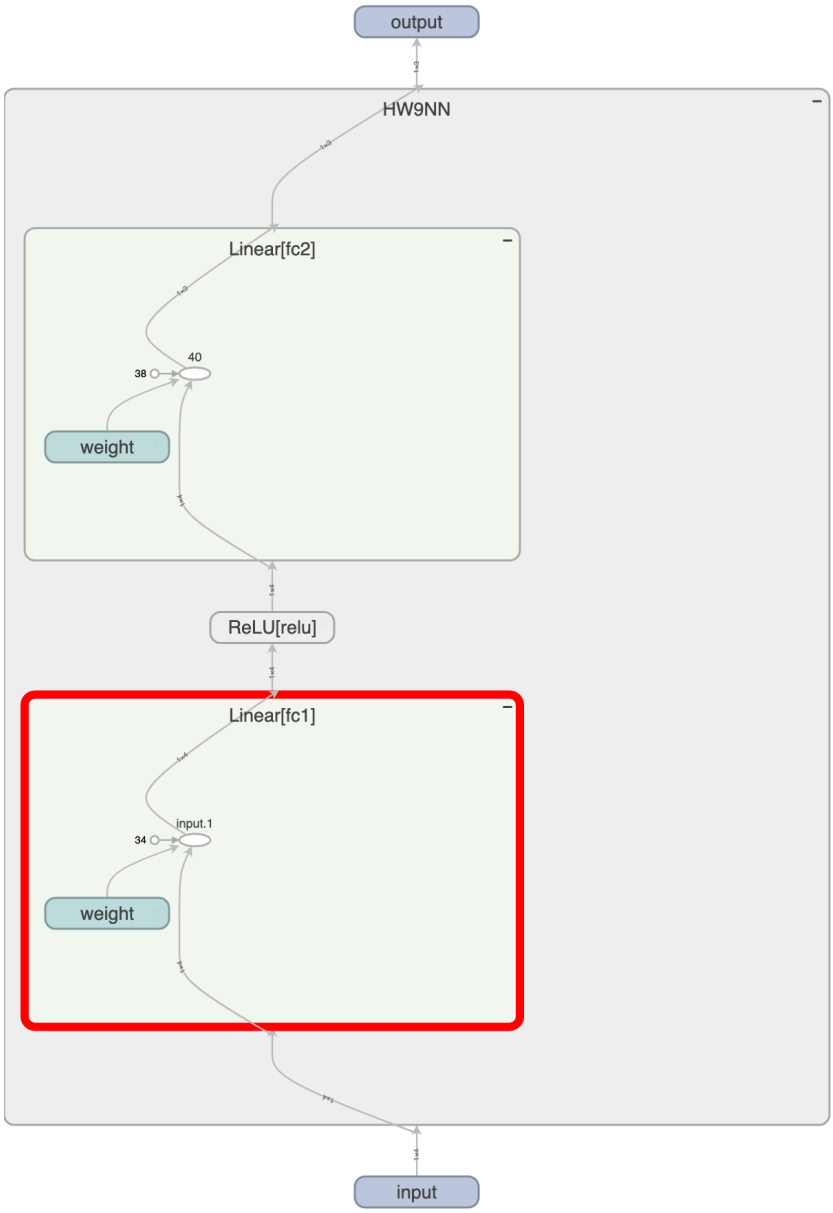
a) Directed Graph



b) Gradient Backprop Computation Graph



c) Tensorboard



Question 2: Backpropagation by Hand

$$x = [0.5, -1.0, 2.0, -0.5]$$
$$\hat{y} = [1.0, -2.0, 0.5]$$
$$W_1 = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 0 \end{bmatrix}, \quad W_2 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

1. Forward Pass

$\boldsymbol{u_1 = xW_1}$

$$u_1 = \begin{bmatrix} 0.5 & -1.0 & 2.0 & -0.5 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 0 \end{bmatrix}$$

- $u_{1,1} = (0.5)(1) + (-1)(0) + (2)(1) + (-0.5)(-1) = 3.0$
- $u_{1,2} = (0.5)(0) + (-1)(1) + (2)(1) + (-0.5)(0) = 1.0$
- $u_{1,3} = (0.5)(-1) + (-1)(0) + (2)(1) + (-0.5)(1) = 1.0$
- $u_{1,4} = (0.5)(0) + (-1)(-1) + (2)(1) + (-0.5)(0) = 3.0$

$$u_1 = [3.0, 1.0, 1.0, 3.0]$$

$\boldsymbol{z_1 = \mathrm{ReLU}(u_1)}$

$$z_1 = [3.0, 1.0, 1.0, 3.0]$$

$\boldsymbol{y = z_1 W_2}$

$$y = \begin{bmatrix} 3.0 & 1.0 & 1.0 & 3.0 \end{bmatrix} \times \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

- $y_1 = 3(1) + 1(0) + 1(1) + 3(-1) = 1.0$
- $y_2 = 3(-1) + 1(1) + 1(0) + 3(1) = 1.0$
- $y_3 = 3(0) + 1(-1) + 1(1) + 3(0) = 0.0$

$$y = [1.0, 1.0, 0.0]$$

$\boldsymbol{\mathcal{L} = \|y - \hat{y}\|_1}$

$$y - \hat{y} = [0.0, 3.0, -0.5]$$
$$\mathcal{L} = |0.0| + |3.0| + |-0.5| = 3.5$$

2. Backward Pass

$\boldsymbol{\frac{\partial \mathcal{L}}{\partial y}}$

$$\frac{\partial \mathcal{L}}{\partial y} = [0, 1, -1]$$

$\boldsymbol{\frac{\partial \mathcal{L}}{\partial W_2} = z_1^T \cdot \frac{\partial \mathcal{L}}{\partial y}}$

$$\frac{\partial \mathcal{L}}{\partial W_2} = \begin{bmatrix} 3 \\ 1 \\ 1 \\ 3 \end{bmatrix} \begin{bmatrix} 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 3 & -3 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 3 & -3 \end{bmatrix}$$

$\boldsymbol{\frac{\partial \mathcal{L}}{\partial z_1} = \frac{\partial \mathcal{L}}{\partial y} \cdot W_2^T}$

$$\frac{\partial \mathcal{L}}{\partial z_1} = \begin{bmatrix} 0 & 1 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 1 & -1 \\ -1 & 1 & 0 & 1 \\ 0 & -1 & 1 & 0 \end{bmatrix}$$

$$= [0, 0, 0, 0] + [-1, 1, 0, 1] - [0, -1, 1, 0] = [-1, 2, -1, 1]$$

$\boldsymbol{\frac{\partial \mathcal{L}}{\partial u_1} = \frac{\partial \mathcal{L}}{\partial z_1} \odot \sigma'(u_1)}$

$$\frac{\partial \mathcal{L}}{\partial u_1} = [-1, 2, -1, 1] \odot [1, 1, 1, 1] = [-1, 2, -1, 1]$$

$\boldsymbol{\frac{\partial \mathcal{L}}{\partial W_1} = x^T \cdot \frac{\partial \mathcal{L}}{\partial u_1}}$

$$\frac{\partial \mathcal{L}}{\partial W_1} = \begin{bmatrix} 0.5 \\ -1.0 \\ 2.0 \\ -0.5 \end{bmatrix} \begin{bmatrix} -1 & 2 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -0.5 & 1.0 & -0.5 & 0.5 \\ 1.0 & -2.0 & 1.0 & -1.0 \\ -2.0 & 4.0 & -2.0 & 2.0 \\ 0.5 & -1.0 & 0.5 & -0.5 \end{bmatrix}$$